Computational Lab Report: Liquid Simulation

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Abstract

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1 Introduction

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Questions:
1. Give a brief account on what molecular dynamics is, and (broadly speaking) what it aims to achieve. [5]
2. Give two examples of instances where molecular dynamics simulations have/can be used (in a helpful way) instead of or in synergy with experimental techniques. In each case explain why. [10] (Please consider only chemically relevant examples. E.g. To study geochemical processes which occur under extreme temperature/pressure conditions, and therefore difficult to study experimentally.)
3. Computer simulations are performed under specific "experimental" conditions. The thermodynamic ensemble defines these conditions. Explain what is meant by the thermodynamic ensemble. Your answer should also provide three ensemble examples and briefly discuss what quantities are conserved in each ensemble. You may want to consult Atkins' Physical Chemistry 11th edition (Focus 13D) to address this question. [5]

2 Results and Discussion

2.1 Introduction to molecular dynamics simulation

TASK 1: By taking Taylor expansions of $x(t+\delta t)$ and $x(t-\delta t)$, write general expressions for them up to the fourth order $\mathcal{O}(t^4)$. Having written these expressions, derive the formula in Figure 2 used to update positions in the classical Verlet algorithm: $x_i(t+\delta t) \approx 2x_i(t) - x_i(t-\delta t) + \frac{F_i(t)}{m}\,\delta t^2$ by using Newton's second law to replace $\frac{d^2x_i(t)}{dt^2}$. [4 marks]

For a function $f(x)$ that is infinitely differentiable at a, Taylor expansion is defined as:

$$f(a+\delta x) = \sum_{n=0}^{+\infty} \frac{f^{(n)}(a)}{n!}(\delta x)^n.$$

Let $x_i(t+\delta t)$ denote the position of an arbitrary atom i at time $t+\delta t$. Expanding $x_i(t+\delta t)$ up to the fourth order gives:

$$x(t+\delta t) = x(t) + x'(t)\delta t + \frac{x''(t)}{2}(\delta t)^2 + \frac{x'''(t)}{6}(\delta t)^3 + \mathcal{O}(t^4).$$

Since $x'(t) = v(t)$ and $x''(t) = a(t)$,

$$x(t+\delta t) = x(t) + v(t)\delta t + \frac{a(t)}{2}(\delta t)^2 + \frac{a'(t)}{6}(\delta t)^3 + \mathcal{O}(t^4).$$

Substituting δt by $-\delta t$, we obtain

$$x(t-\delta t) = x(t) - v(t)\delta t + \frac{a(t)}{2}(\delta t)^2 - \frac{a'(t)}{6}(\delta t)^3 + \mathcal{O}(t^4).$$

Adding $x(t-\delta t)$ to $x(t+\delta t)$ gives

$$x(t+\delta t) + x(t-\delta t) = 2x(t) + a(t)(\delta t)^2 + \mathcal{O}(t^4).$$

Using Newton's law ($F(t) = ma(t)$) and omitting $\mathcal{O}(t^4)$, the equation becomes

$$x(t+\delta t) + x(t-\delta t) \approx 2x(t) + \frac{F(t)}{m}(\delta t)^2.$$

By rearranging, we arrive at

$$x(t+\delta t) \approx 2x(t) - x(t-\delta t) + \frac{F(t)}{m}(\delta t)^2,$$

which is the formula used in the classic Verlet algorithm (CVA, [1]).

TASK 2: What could be an advantage of the Velocity-Verlet algorithm over the classical Verlet algorithm? [2 marks]

The Velocity-Verlet algorithm (VVA, [2]) lets us calculate the atom velocities explicitly by using

$$v_i(t+\delta t) = v_i(t+\frac{1}{2}\delta t) + \frac{1}{2}a_i(t+\delta t)\delta t.$$

Though when VVA is adopted, we need to specify the initial velocities apart from the initial positions, this algorithm comes with several advantages compared to the CVA. A number of physical, especially dynamical quantities depend on velocity, e.g., kinetic energy, temperature, where VVA allows accurate calculation of these quantities at each timestep without approximation. Additionally, while retaining the same global error, stochastic collisions of atoms can easily be implemented for the VVA algorithm as velocities are computed for the entire trajectory. [3]

TASK 3: Consider the Lennard-Jones pair potential. What physical interaction(s) does it describe? What is the physical significance for the r^{-6} and r^{-12} terms? [3 marks]

The Lennard–Jones (LJ) pair potential describes the interaction between two neutral atoms or molecules through a combination of attractive and repulsive forces. It was first proposed in 1924 to describe the cohesive energy of crystals of noble gases, [4] and has since been widely used to model intermolecular interactions in simple liquids and gases. [5]
The attractive r^{-6} term represents long-range London dispersion (van der Waals) interactions arising from instantaneous dipole–induced dipole correlations between particles. The repulsive r^{-12} term models the strong short-range repulsion that occurs when electron clouds overlap, which is primarily a consequence of the Pauli exclusion principle. The behaviours of the potential from the two terms at different distances are shown in Fig. 1.
The depth of the potential well, ε, represents the strength of the interaction between two particles and corresponds to the minimum potential energy at the equilibrium separation. Physically, it reflects the cohesive energy scale of the interaction and determines how strongly particles are bound in the system.

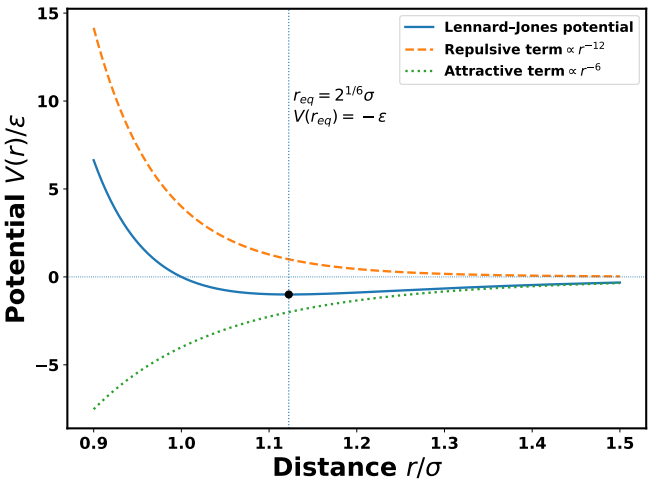


Figure 1: Lennard–Jones potential showing the short-range repulsive r^{-12} term and long-range attractive r^{-6} dispersion interaction. The potential minimum occurs at $r_{eq} = 2^{1/6}\sigma$, where the energy is $V(r_{eq}) = -\varepsilon$. The well depth ε represents the interaction strength between two particles.

TASK 4: For a single Lennard–Jones interaction, $\phi(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$, find the separation r_0 at which the potential energy is zero. What is the force at this separation? Find the equilibrium separation r_{eq} and determine the well depth $\phi(r_{\text{eq}})$. Evaluate the integrals $\int_{2\sigma}^{\infty}\phi(r)\,dr$, $\int_{2.5\sigma}^{\infty}\phi(r)\,dr$, and $\int_{3\sigma}^{\infty}\phi(r)\,dr$ when $\sigma = \varepsilon = 1.0$. [4 marks]

Let $\phi(r) = 0$, dividing both sides by 4ε and moving repulsive term to the RHS, we obtain

$$\sigma^{12}r^{-12} = \sigma^{6}r^{-6}.$$

Simple algebra gives

$$r^{-6} = \sigma^{-6}$$

and therefore $r_0 = \sigma$, given both r and σ are positive.
The force at r_0 can be evaluated by taking the derivative of the LJ potential at r_0:

$$\begin{aligned}\frac{d\phi(r)}{dr} &= 4\varepsilon\left(\sigma^{12}\cdot -12r^{-13} - \sigma^{6}\cdot -6r^{-7}\right)\\ &= 4\varepsilon\left(-12\sigma^{12}r^{-13} + 6\sigma^{6}r^{-7}\right).\end{aligned}$$

At $r = r_0$,

$$\begin{aligned}\frac{d\phi(r)}{dr}\Big|_{r=\sigma} &= 4\varepsilon\left(-12\sigma^{-1} + 6\sigma^{-1}\right)\\ &= -24\varepsilon\sigma^{-1}.\end{aligned}$$

Therefore, the force at r_0 is

$$F(r_0) = -\frac{d\phi(r)}{dr}\Big|_{r=\sigma} = 24\varepsilon\sigma^{-1},$$

indicating a repulsive force at $r_0 = \sigma$.
At the equilibrium separation (r_{eq}), the derivative of the LJ potential, i.e., the force is 0. To find r_{eq}, we can solve the equation

$$F = -4\varepsilon\left(-12\sigma^{12}r_{\text{eq}}^{-13} + 6\sigma^{6}r_{\text{eq}}^{-7}\right) = 0.$$

Dividing both sides by 4ε and rearranging the equation gives

$$6\sigma^{6}r_{\text{eq}}^{-7} = 12\sigma^{12}r_{\text{eq}}^{-13}.$$

With simple algebra, we can find that

$$r_{\text{eq}} = (2)^{1/6}\sigma,$$

Substituting $r = r_{\text{eq}}$ into the LJ potential gives

$$\begin{aligned}\phi(r_{\text{eq}}) &= 4\varepsilon\left[\left(\frac{\sigma}{r_{\text{eq}}}\right)^{12} - \left(\frac{\sigma}{r_{\text{eq}}}\right)^{6}\right]\\ &= 4\varepsilon\left[\left(\frac{\sigma}{(2)^{1/6}\sigma}\right)^{12} - \left(\frac{\sigma}{(2)^{1/6}\sigma}\right)^{6}\right]\\ &= 4\varepsilon\left(\left(2^{-1/6}\right)^{12} - \left(2^{-1/6}\right)^{6}\right)\\ &= 4\varepsilon\cdot -2^{-2}\\ &= -\varepsilon.\end{aligned}$$

Let $\sigma = \varepsilon = 1.0$, the LJ potential becomes

$$\phi(r) = 4r^{-12} - 4r^{-6}.$$

The integral of this LJ potential with infinity as upper limit and a (where $a > 0$) as lower limit is

$$\begin{aligned}\int_{a}^{+\infty}\phi(r)\,dr &= \left[-\frac{4}{11}r^{-11} + \frac{4}{5}r^{-5}\right]_{a}^{+\infty}\\ &= 0 - \left[-\frac{4}{11}a^{-11} + \frac{4}{5}a^{-5}\right]\\ &= \frac{4}{11}a^{-11} - \frac{4}{5}a^{-5}.\end{aligned}$$

The integrals can then be evaluated as below:

$$\int_{2\sigma}^{\infty}\phi(r)\,dr = \frac{4}{11}\cdot 2^{-11} - \frac{4}{5}\cdot 2^{-5} \approx -0.0248 \text{ (to 3 s.f.)}$$
$$\int_{2.5\sigma}^{\infty}\phi(r)\,dr = \frac{4}{11}\cdot 2.5^{-11} - \frac{4}{5}\cdot 2.5^{-5} \approx -0.00818 \text{ (to 3 s.f.)}$$
$$\int_{3\sigma}^{\infty}\phi(r)\,dr = \frac{4}{11}\cdot 3^{-11} - \frac{4}{5}\cdot 3^{-5} \approx -0.00329 \text{ (to 3 s.f.)}$$

From these results, the magnitude of $\int_{a}^{\infty}\phi(r)\,dr$ decreases as the lower limit a increases. This shows that the long-range contribution of the LJ potential becomes progressively smaller at larger separations, consistent with the fact that $\phi(r) \to 0$ as $r \to +\infty$.

TASK 5: Consider an atom at position (0.5, 0.5, 0.5) in a cubic simulation box that spans from (0, 0, 0) to (1, 1, 1). In a single timestep, it moves along the vector (0.7, 0.6, 0.2). At what position does the atom end up after applying periodic boundary conditions? [1 mark]

After the atom moves, its position becomes

$$(0.5,\ 0.5,\ 0.5) + (0.7,\ 0.6,\ 0.2) = (1.2,\ 1.1,\ 0.7).$$

As the atom crosses the boundary of the simulation box, periodic boundary conditions map its position back into the box by subtracting the box length in the corresponding directions, and therefore, the atom ends up at

$$(0.2,\ 0.1,\ 0.7).$$

TASK 6: The Lennard–Jones parameters for argon are $\sigma = 0.34\,\mathrm{nm}$ and $\varepsilon/k_B = 120\,\mathrm{K}$. If the reduced LJ cutoff is $r^* = 3.2$, what is it in real units? What is the well depth in $\mathrm{kJ\,mol^{-1}}$? What is the reduced temperature $T^* = 1.5$ in real units? [1 mark]

The reduced units in LJ system adopt σ as unit of length, ε as unit of energy, and m as unit of mass (where m is the mass of the atoms of interest in the system) [6]. The "real" units and reduced units can be linked by:

$$r \equiv r^*\sigma, \qquad T = T^*\frac{\varepsilon}{k_B}.$$

Therefore, the LJ cutoff in real unit is

$$r = r^*\sigma = 3.2 \cdot 0.34 \text{ nm} = 1.088 \text{ nm}.$$

The well depth in J is

$$\begin{aligned}\varepsilon(\mathrm{J}) &= k_B \cdot 120\,\mathrm{K} = 1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \cdot 120\,\mathrm{K} \\ &= 1.6572 \times 10^{-21}\,\mathrm{J}.\end{aligned}$$

To obtain the well depth in $\mathrm{kJ\,mol^{-1}}$, we can multiply ε by Avogadro's number and converting J to kJ:

$$\begin{aligned}\varepsilon(\mathrm{kJ\,mol^{-1}}) &= \varepsilon(\mathrm{J}) \cdot 10^{-3}\ \mathrm{kJ\ J^{-1}} \cdot N_A \\ &= 1.6572 \times 10^{-24}\,\mathrm{kJ} \cdot 6.022 \times 10^{23}\ \mathrm{mol^{-1}} \\ &\approx 0.997\ \mathrm{kJ\ mol^{-1}}\ \text{(to 3 s.f.)}.\end{aligned}$$

And the temperature in Kelvin is

$$\begin{aligned}T(\mathrm{K}) &= T^*\frac{\varepsilon}{k_B} \\ &= 120 \cdot 1.5\ \mathrm{K} \\ &= 180\ \mathrm{K}.\end{aligned}$$

2.2 Equilibration

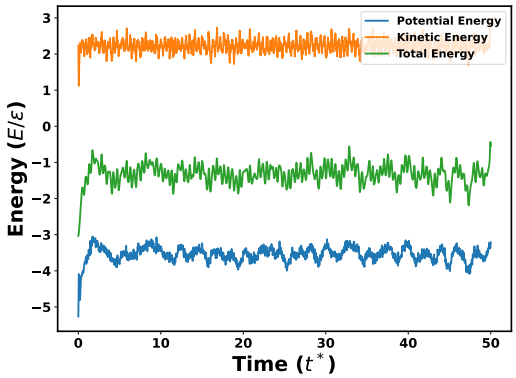
What does it mean for a simulation to "reach equilibrium"? Why is this important in terms of sampling from an ensemble using molecular dynamics? [2]

The simulated system reaches equilibrium when its macroscopic quantities (i.e., temperature, pressure and energy) fluctuate around the stable average values. At this stage, the system has reached a steady thermodynamic state consistent with the specified ensemble.
This is important because MD samples configurations in a statistical way in the ensemble only after the system has equilibrated. If the simulation has not reached equilibrium, the sampled configurations may still reflect the initial conditions rather than the true equilibrium distribution, leading to incorrect averages of physical properties.

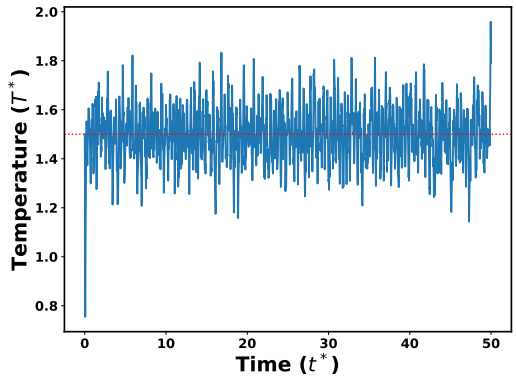
Does the simulation reach equilibrium? How can you tell? [2]

The simulation has reached equilibrium. From Fig. 2, the thermodynamic quantities fluctuate around approximately constant mean values without systematic drift. In particular, the temperature and pressure fluctuate around the target reduced values ($T^* = 1.5$, $P^* = 1$) specified in the input file. The potential, kinetic, and total energies also exhibit stationary fluctuations after a short initial relaxation period. These observations indicate that the system has equilibrated.

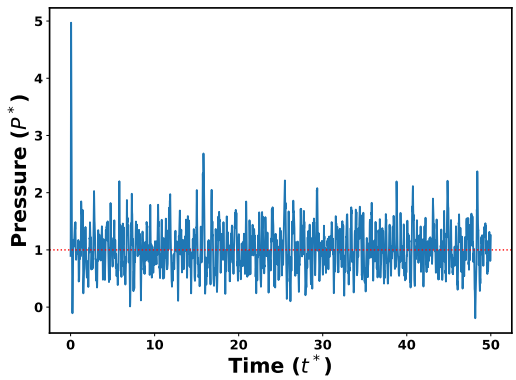
Plot the energy (potential, kinetic and total), temperature and pressure, against time for the 0.001 timestep experiment [2]



(a) Potential, kinetic, and total energy vs time.



(b) Temperature vs time ($T^* = 1.5$).



(c) Pressure vs time ($P^* = 1$).

Figure 2: Thermodynamic properties of the system during the NPT simulation with timestep 0.001, where T^* and P^* is set to 1.5 and 1, respectively.

Make a single plot which shows the energy vs. time for the timesteps you have simulated [2].

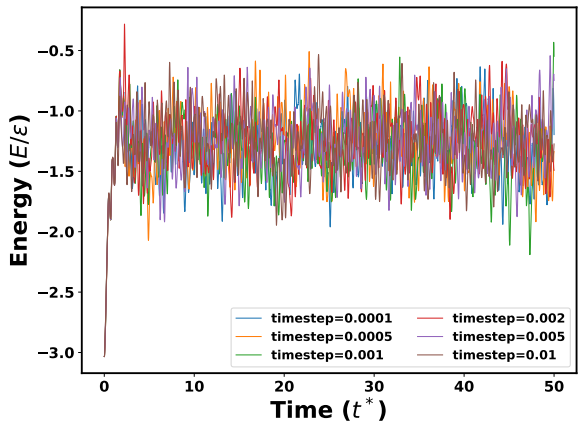


Figure 3: Total energy as a function of simulation time for NPT simulations with different timesteps. After equilibration, the energy fluctuates around a stable mean value for all tested timesteps, indicating numerical stability.

Of the timesteps that you used, which timestep will you use for subsequent simulations and why? [6]

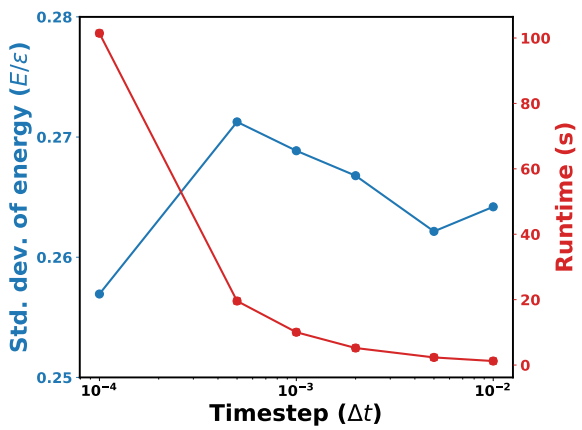


Figure 4: Standard deviation of the total energy (blue, left axis) and simulation runtime (red, right axis) as a function of integration timestep. The energy fluctuations remain similar across the tested timesteps, indicating numerical stability, while the computational runtime decreases rapidly with increasing timestep.

The standard deviation of the energy and total simulation runtime for the simulation with different timesteps are shown in Fig. 4 (with x-axis plotted in log scale). It is found that while standard deviation remains similar, the total runtime decreases approximately inversely proportional to the timestep. Timestep $\tau = 0.001$ was therefore selected for subsequent simulations, as it provides good compromise between numerical stability and computational efficiency. Reducing the timestep further would only give minimal improvement with substantial increase in the runtime.

Physically, the timestep is the internal of physical time over which the equations of motion are integrated during the simulation. The unit of timestep τ is defined as: [6]

$$\tau = \sqrt{\frac{m\sigma^2}{\varepsilon}}.$$

At each timestep, the forces of the atoms are calculated, and positions and/or velocities are updated discretely. τ represents the time between two successive updates of the state of the system. Hence if the timestep is too large, the force calculations would become inaccurate, and the simulation is no longer valid physically. This effect is especially significant for LJ system, as it has a very strong repulsive term at short distances (e.g., at $r = 0.9\ \sigma$, the LJ potential is already $5\ \varepsilon$). Therefore, the timestep must be small enough to *resolve* the motions of the atoms.
On the other hand, if the timestep is too small, though the integration becomes highly accurate and the system becomes stable, the simulation becomes very compuatationally expensive.

2.3 Running simulations under specific conditions

Plot the density as a function of temperature for both pressures that you simulated. Include a line corresponding to the predictions made by the ideal gas law. [3]

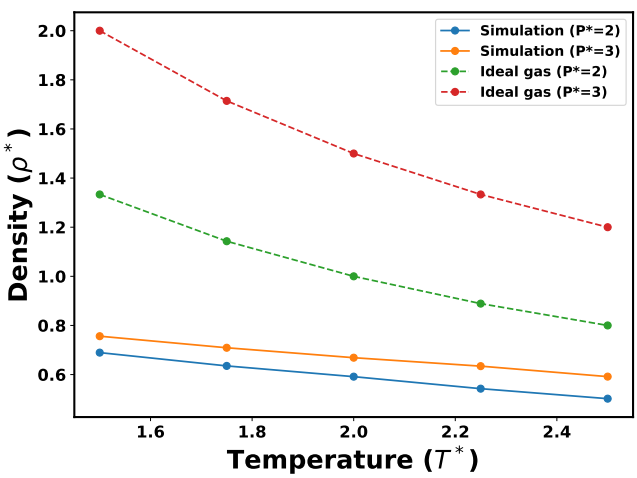


Figure 5: Density as a function of temperature for LJ simulations at pressures $P^* = 2$ and 3 and temperatures $T^* = 1.5,\ 1.75,\ 2,\ 2.25,\ 2.5$. The simulated densities are significantly lower than those predicted by the ideal gas law due to intermolecular interactions and excluded volume effects present in the LJ potential.

How do your results compare to the ideal gas law? Do deviations increase/decrease with temperature and pressure? Explain. [7]

Ideal gas law gives $PV = Nk_BT$ and therefore

$$\rho = \frac{P}{k_BT}.$$

Using the reduced unit, where $P^* = P\sigma^3\varepsilon^{-1}$, $T^* = k_B T\varepsilon^{-1}$, and $\rho^* = \rho\sigma^3$, this becomes

$$\rho^* = \frac{P^*}{T^*}.$$

Compared with the ideal gas prediction, the simulations yield significantly lower densities at the same temperature and pressure. This deviation arises because the ideal gas law neglects intermolecular interactions and assumes point-like particles, whereas the LJ potential includes both repulsive and attractive interactions. The repulsive core introduces excluded volume effects, while the attractive interactions reduce the effective pressure of the system compared with an ideal gas.
As the temperature increases, the deviation from ideal-gas behaviour decreases. At higher temperatures the kinetic energy dominates over the interparticle potential energy, so the system behaves more like an ideal gas.
In contrast, the deviation increases with pressure. At higher pressure the particles are forced closer together, making intermolecular interactions and excluded volume effects more significant.
This trend is consistent with the results shown in Fig. 6, where the deviation is largest at low temperature and high pressure.

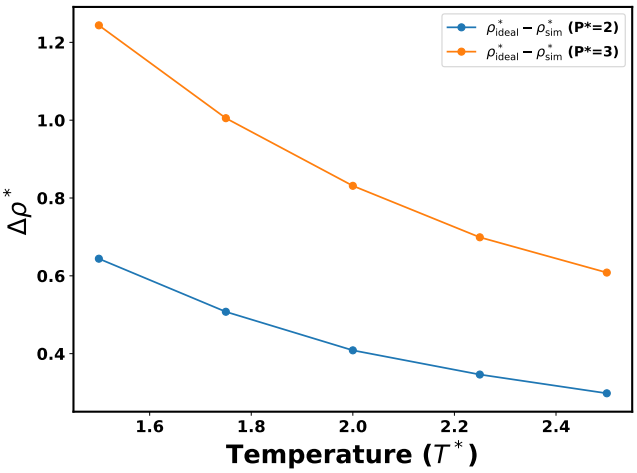


Figure 6: Deviation of the simulated density from the ideal-gas prediction as a function of temperature at two pressures. The deviation decreases with increasing temperature and increases with pressure.

Do you expect your simulation results to be in better or worse agreement with the Van der Waals equation of state? Why? [3]

The Van der Waals equation of state has the form

$$(P + \frac{aN^2}{V^2})(V - Nb) = RT,$$

where a is attraction parameter, which corrects the attractive forces between particles, and b is the excluded volume parameter, which corrects the finite size of molecules.
Compared to the ideal gas law, the two parameters introduced in the equation take the potential energy and particle volume into account, thereby the deviations could be reduced to a large extent, resulting in better agreement with MD simulation using LJ potential.

2.4 Structural properties and the radial distribution function

Briefly explain what a Radial Distribution Function is. What is the relationship between the coordination number and RDF? [2]

Radial distribution function (RDF, $g(r)$) describes the probability of finding a particle at a distance r from a reference particle relative to that of an ideal gas at the same density. It therefore characterises the spatial correlations and local structure of a system.
The coordination number (n) indicates the average number of neighbouring particles that can be found within the first coordination shell of the reference particle. Mathematically, it is obtained by integrating the RDF as [7]:

$$n = 4\pi\rho \int_0^{r_{\min}} r^2 g(r)\ dr,$$

where ρ is the average density of the particles and $r_{\min}$ is the position of first minimum of $g(r)$.

Plot the RDFs for the three systems. [2]

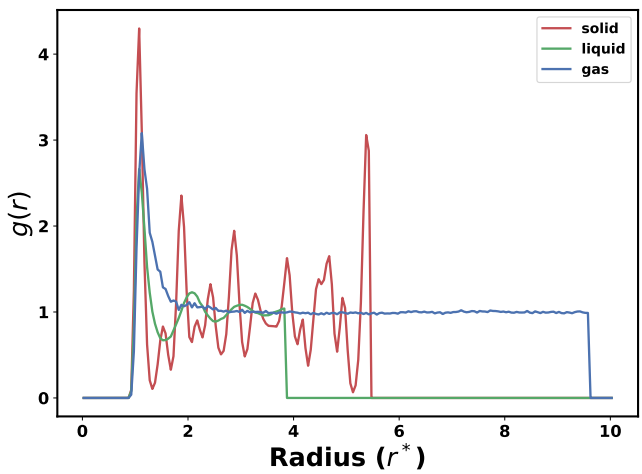


Figure 7: Radial distribution functions $g(r)$ for the solid (FCC), liquid, and gas phases of the LJ system.

Discuss qualitatively the differences between the three RDFs, and what this tells you about the structure of the system in each phase. [5]

The RDF of the solid phase displays multiple sharp peaks up to approximately $r^* = 6$, reflecting the long-range periodic structure of the FCC lattice.
The liquid phase exhibits a strong first peak and damped oscillations corresponding to short-range order.
It is found that both the solid and liquid phases are cut off at a certain radius, illustrating the finite size of the simulation box and the maximum distance over which the radial distribution function is computed.

The gas phase RDF has a similar first peak to those of solid and liquid phase, after which it approaches $g(r) = 1$, consistent with negligible spatial correlations.

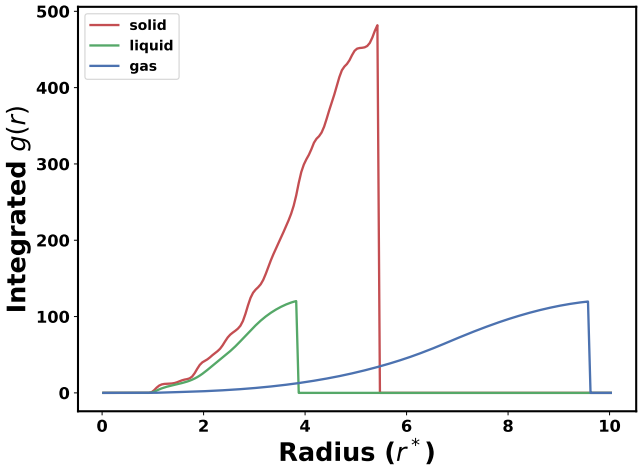


Figure 8: Integrated radial distribution functions for the solid (FCC), liquid, and gas phases of the LJ system. The integrated $g(r)$ represents the cumulative number of particles within a distance r from a reference particle.

Quantitative analysis of the RDF curves reveals the value of first $r_{\min}$ (i.e., first coordination sphere) and the coordination number of each state (Table 1). Simulation of the FCC solid yields a coordination number of 12.08, in excellent agreement with the theoretical value of 12. A similar coordination number with larger first coordination sphere is found for the liquid phase simulation, indicating the retention of short-range packing to a large extent. However, from Fig. 7, the damped oscillations after the first $r_{\min}$ demonstrates the lack of long-range order. The first $r_{\min}$ for gas is the highest among all, with significantly lower coordination number. This is consistent with the much lower density of the system, aligning with the weak and broad RDF curve and the nature of gas. Overall, the RDF clearly distinguishes the structural characteristics of the three phases, where the progressive reduction in structural order from solid to liquid to gas is directly reflected in the corresponding RDF profiles.

Table 1: First minimum of the radial distribution function and corresponding coordination numbers for the three phases.

	Solid	Liquid	Gas
First $r_{\min}$	1.33	1.53	1.68
n	12.08	12.05	1.39

In the solid case, illustrate which lattice sites the first three peaks correspond to [2]. What is the lattice spacing [1]? What are the coordination number for each of the first three peaks [1]?

Let a denote the cubic lattice parameter, then the first three peaks, i.e., the first three coordination shell distances are:

$$r_1 = \frac{a}{\sqrt{2}}, \qquad r_2 = a, \qquad r_3 = \sqrt{\frac{3}{2}}a.$$

The the corresponding lattice sites for the first peak include:

$$\left(0, \frac{a}{2}, \frac{a}{2}\right), \qquad \left(\frac{a}{2}, 0, \frac{a}{2}\right), \qquad \left(\frac{a}{2}, \frac{a}{2}, 0\right),$$

and symmetry-equivalent points. The coordination number of the first peak is 12. The lattice spacing can be obtained from the first peak position as

$$a = \sqrt{2}\, r_1.$$

The the corresponding lattice sites for the second peak include:

$$(a, 0, 0), \qquad (0, a, 0), \qquad (0, 0, a),$$

and symmetry-equivalent points. The coordination number of the second peak is 6.
The the corresponding lattice sites for the third peak include:

$$\left(a, \frac{a}{2}, \frac{a}{2}\right), \qquad \left(\frac{a}{2}, a, \frac{a}{2}\right), \qquad \left(\frac{a}{2}, \frac{a}{2}, a\right),$$

and symmetry-equivalent points. The coordination number of the third peak is 24.

Table 2: Radial positions of the first three RDF shell minima and the corresponding coordination numbers for the FCC solid from the simulation.

	First shell	Second shell	Third shell
$r_{\min}$	1.33	1.68	2.08
n	12.08	5.95	25.91

From Table 2, it is found that the simulation reproduces the expected coordination numbers of an FCC lattice. The first three coordination shells yield approximately 12.08, 5.95, and 25.91 neighbours, which are in good agreement with the theoretical values of 12, 6, and 24 for an ideal FCC structure.

2.5 Dynamical properties and the diffusion coefficient

3 Conclusions

Conclusions Conclusions Conclusions
Questions
1. In this lab, you have used the Lennard-Jones potential exclusively. Give one example of a system that can be described well using only LJ potentials. What additional elements would you need to add to the interaction potential to model a molecule, e.g. water? [6]
2. What cut-off have you been using in your simulations? What are the advantages and disadvantages for using a shorter/longer cut-off? (Hint: consider here the balance of computational cost vs accuracy in describing the properties of a system.) [3]
3. In this lab you have investigated the properties of solids, liquids and gases. Would you observe a liquid phase (and by extension a critical point) if the LJ interaction strength, ε, is very weak? What interaction strength is required to

generate a liquid phase? (Hint: to address this question you may want to revise the 2nd year Solids, Liquid and Interfaces lecture notes - see "Phase transitions of pure substances") [4]
4. What are finite size effects? Do you think they are significant in the simulations you have performed? Why? [3]
5. Algorithms such as SHAKE and RATTLE are widely used to model molecular systems as they allow MD simulations to be performed while fixing bond lengths. Why is this desirable? [4] (Hint: think about the timestep.)

References

[1] Verlet, L. Computer" experiments" on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules. *Physical review* **1967**, *159*, 98.

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Supporting Information